PLUREL: Synthetic Data unlocks Scaling Laws for Relational Foundation Models

March 2, 2026

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[1] Kothapalli, Vignesh, et al. "PluRel: Synthetic Data unlocks Scaling Laws for Relational Foundation Models." arXiv preprint arXiv:2602.04029 (2026).

Motivation

Learning from **multi-table relational databases**
E-commerce example:

Users

user_id	age	country
1	22	US
2	35	UK
3	29	CN

Orders

order_id	user_id	item_id	price
101	1	11	19.9
102	1	15	5.2
103	2	11	19.9

Items

item_id	category	stock
11	book	200
15	food	80
21	toy	40

- User churn prediction
- Purchase propensity prediction
- ...

We use Relational Foundation Models (RFMs) to learn on relational databases and perform numerous predictive tasks.

Motivation

However, large-scale multi-table relational databases are rarely publicly available due to privacy and proprietary business constraints.

Goal: **PLUREL** is used to **synthesize** realistic relational databases for pretraining, providing a scalable data source for next-generation Relational Foundation Models (RFMs).

Method

Basic Notions:

- **Primary key**: a column that uniquely identifies each row in a table
- **Foreign key**: a column whose value references the primary key of another table
- **Features**: attribute columns describing properties of an entity or an interaction

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Method

- **Schema**: database structure and table relations
- **Row-level Connectivity**: links rows across tables via keys
- **Feature**: patterns of attribute values in tables

Schema Generation

- Sample $\mathcal{G}$ from a family of random directed acyclic graphs
- Each node represents a table with a primary key and associated features
- Child nodes depend on their parent nodes through primary-foreign key links.
- The number of rows and feature columns is sampled randomly

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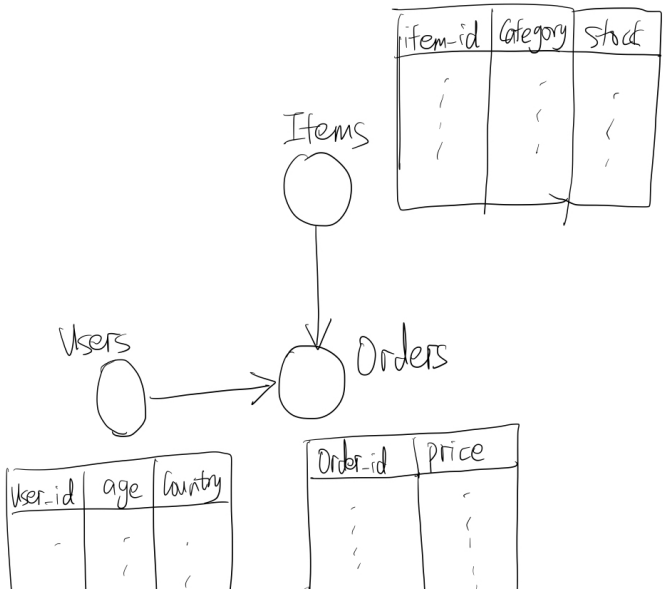
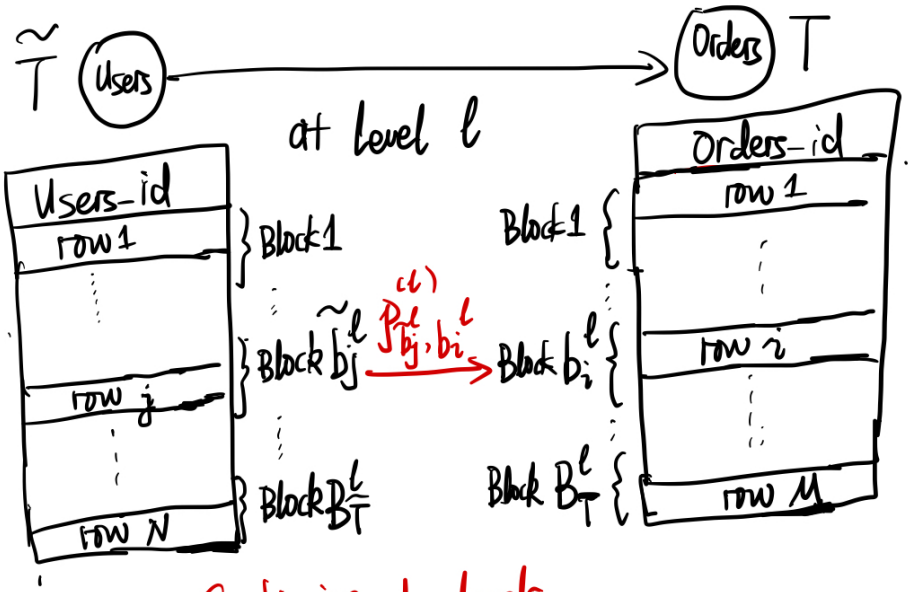


Row-level Connectivity Generation

- Consider a node T with parent $\tilde{T}$. We cluster rows of the two tables into groups and use a Hierarchical Stochastic Block Model (HSBM) to generate their connections.
- HSBM: Rows in each table are partitioned into L hierarchical blocks.



Combining L levels,

$$p(j \to i) = \frac{s_{ij}}{\sum_{k=1}^{M} s_{ik}}, \quad s_{ij} = \prod_{\ell=1}^{L} p^{(\ell)}_{\tilde{b}_j^\ell, b_i^{(\ell)}}$$

Row-level Connectivity Generation:HSBM

Hierarchical grouping. Rows in each table are partitioned into L hierarchical blocks

$$H_T = (B_T^1, \ldots, B_T^L), \qquad H_{T'} = (B_{T'}^1, \ldots, B_{T'}^L).$$

Each row receives a block vector:

$$\mathbf{b}_i = (b_i^1, \ldots, b_i^L) \ \ (i \in T'), \qquad \tilde{\mathbf{b}}_j = (\tilde{b}_j^1, \ldots, \tilde{b}_j^L) \ \ (j \in T).$$

Block interaction probabilities-HSBM At every level ℓ, connectivity is controlled by a matrix

$$P^{[\ell]} \in \mathbb{R}^{B_{T'}^{\ell} \times B_T^{\ell}},$$

which determines how likely blocks of T connect to blocks of T'.
Foreign-key sampling

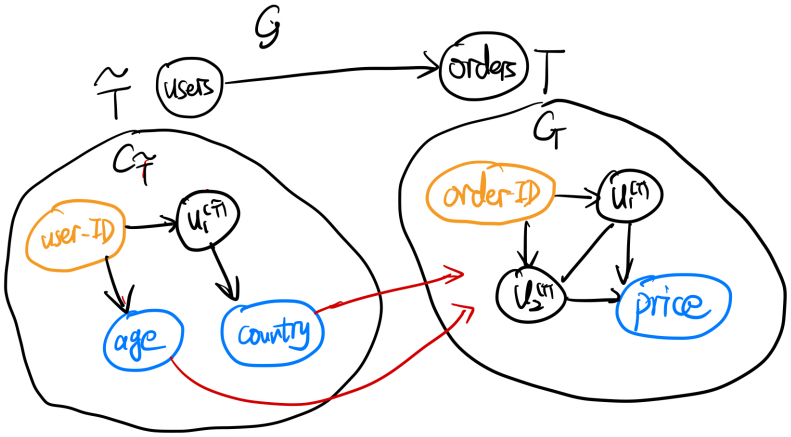
$$P(j \to i) = \frac{s_{ij}}{\sum_{k=1}^{M} s_{kj}}, \text{ with } s_{ij} = \prod_{\ell=1}^{L} P^{[\ell]}_{\tilde{b}_j^{\ell}, b_i^{\ell}}.$$

Feature Generation: Structural Causal Models (SCM)

- Source variables (primary key) are sampled and propagated through a causal graph ($C_T = (V_T, E_T)$), and the resulting feature values depend both on internal causal relations and on features from parent tables.
- The mechanism of each node in C_T satisfies

$$z_i = H_i\left(\bigcup_{T' \in \mathrm{Pa}(T,\mathcal{G})} V^F_{T'}, \mathrm{Pa}(v_i, C_T),\ u_i\right).$$

where $V^F_{T'}$ means the nodes of features in $C_{T'}$, $\mathrm{Pa}(T, \mathcal{G})$ represents the parents node of T, and u_i is latent factors (trend, cyclical, and fluctuation).



Experiment Overview

Training	Testing	Goal	Main result
Synthetic	Synthetic	In-distribution scaling	Loss follows power-law scaling vs. #DBs (diversity) and tokens (size).
Synthetic	Real	OOD transfer	Higher synthetic diversity improves zero-shot performance on RelBench.
Synthetic + Real	Real	Task alignment	Best overall: continued pretraining improves up to +7.4% AUROC, +5.2% R^2.

Scaling Laws for pretraining loss

Two scaling axes

- N: number of synthetic relational databases (diversity)
- S: total number of pretraining tokens (data size)

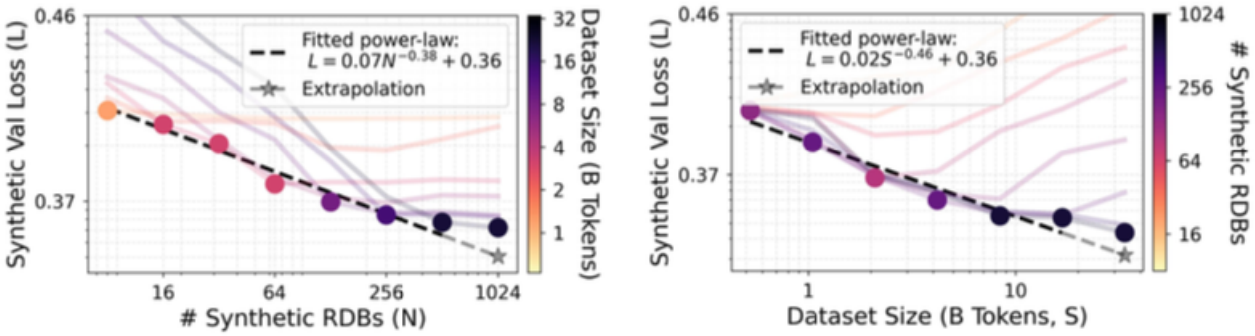
Validation loss

$$L(N, S) = \text{validation loss on held-out synthetic databases}$$

$$L(N) = \min_S L(N, S), \qquad L(S) = \min_N L(N, S).$$

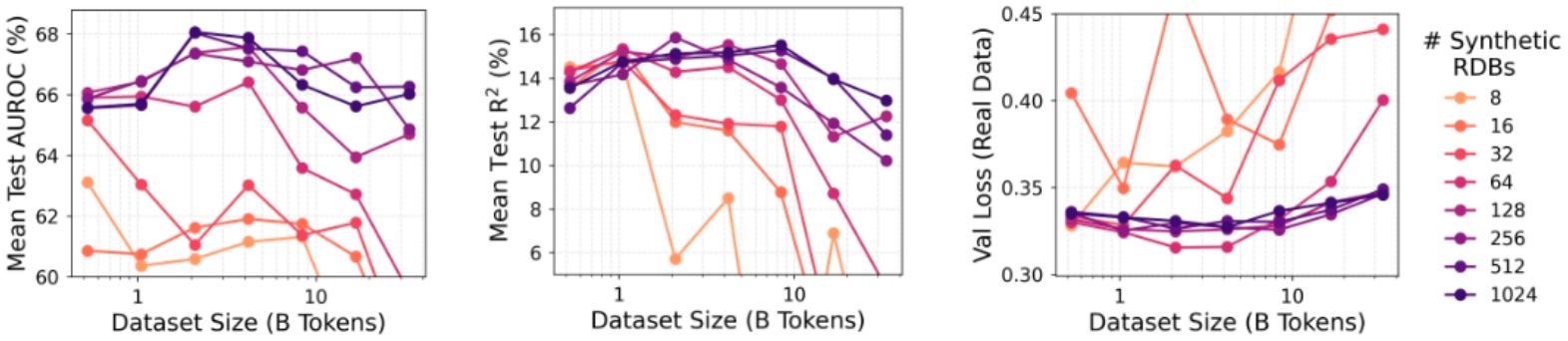
Interpretation

- $L(N) = A_N N^{-\alpha_N} + C_N$ (diversity scaling)
- $L(S) = A_S S^{-\alpha_S} + C_S$ (data size scaling)
- Best performance requires scaling both together



Generalization to Real Datasets

- When diversity is low, adding more tokens alone is ineffective and can even worsen performance.
- Once the number of synthetic databases increases, scaling data size begins to consistently improve results.



(a) Mean 0-shot test AUROC (%) (↑) *(b)* Mean 0-shot test R^2 (%) (↑) *(c)* Validation loss (↓) on real data (RelBench)

Continued Pretraining on Real Datasets

Dataset	Task	Real only	Synthetic + Real (ours)	Absolute Gain (%)	Synthetic only (ours)
AUROC(%) for classification. Higher is better. Majority baseline is 50.0.					
rel-amazon	user-churn	64.2	**65.0**	+0.8	64.4
rel-hm	user-churn	**67.4**	66.0	−1.4	63.7
rel-stack	user-badge	80.0	**82.0**	+2.0	81.4
rel-stack	user-engage	78.9	**86.2**	+7.4	82.4
rel-amazon	item-churn	67.6	**72.5**	+4.9	71.0
rel-avito	user-visits	57.2	**63.4**	+6.2	63.5
rel-avito	user-clicks	**54.7**	47.9	−6.8	45.9
rel-trial	study-out	**54.4**	51.8	−2.6	53.8
rel-f1	driver-dnf	80.7	**81.0**	+0.3	76.7
rel-f1	driver-top3	86.9	**88.4**	+1.5	82.6
Mean		69.2	**70.4**	+1.2	68.5
R^2(%) for regression. Higher is better. Mean baseline is 0.0.					
rel-hm	item-sales	16.0	**20.0**	+4.0	4.4
rel-amazon	user-ltv	14.5	**18.5**	+4.0	9.8
rel-amazon	item-ltv	35.3	**40.5**	+5.2	10.7
rel-stack	post-votes	22.3	**25.5**	+3.2	15.7
rel-trial	site-succ	33.7	**38.6**	+5.0	38.3
rel-trial	study-adv	**1.9**	1.6	−0.3	−0.8
rel-f1	driver-pos	54.3	**55.5**	+1.2	41.3
rel-avito	ad-ctr	3.1	**4.9**	+1.9	2.5
Mean		22.6	**25.7**	+3.0	15.2